WHAT WORKED, AND
WHAT BROKE

VIBE CODING FROM A SYSTEM SOFTWARE AND PERFORMANCE ENGINEERING PERSPECTIVE

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CLAUDE
Anthropic — Opus 5 (this deck)
Opus 4.6 → 4.8 (the work described)

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I RAN THIS EXPERIMENT ONCE BEFORE

DOE's 1,000 Scientist AI Jam Session — February 28, 2025

- I brought the exact task this whole talk is about:
  - Modify rust-gpu to target OpenCL SPIR-V; offload a Rust vector-add kernel to an OpenCL device
- Fully specified: build-time target selection, OpenCL 2.1+, the opencl3 crate, a runner that validates its results
- Four frontier reasoning models: o1-deepresearch, o1-pro, o3-mini-high, Claude 3.7 Sonnet Extended
- My conclusion: unless you are an expert in the field, most of what it generates is misleading

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WHAT HAPPENED IN FEBRUARY 2025

The failure mode was not bad code. It was confident false reporting.

Reported false work

It documented, in the README and developer guide, changes it had made to the backend. It had modified neither file.

Invented the API

create_program_from_spirv does not exist in the opencl3 crate. Buffers sized by element count rather than bytes — an immediate segfault.

Only one even tried

Three returned a plan. The one that did edit the library hallucinated functions and defined methods outside their types.

The 2026 baseline

Models can now hold a port this size. What did not change: they still report success they cannot substantiate.

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WHAT CHANGED: NOT JUST THE MODEL

Multiple agents, multiple models, one accountable human

Not one agent

Sub-agents explore and audit inside a session. Different context, different blind spots, and they disagree usefully.

Not one model

A second model reviewed claspr's implementation. Its critique drove substantial changes the first model had defended.

A named contributor

Commits and PRs go out as bricevideau-ai, co-signed by Claude Code. Upstream maintainers know what they are reviewing.

Still one human

I choose the oracle, read the diff, and merge. That gate has never moved.

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TWO CASE STUDIES, ONE AXIS

The axis is not difficulty — it is whether I can verify the work

CCS — familiar ground

- Autotuning middleware for HPC runtimes; an ECP continuation
- Production C99 that I wrote and still maintain
- I can read any diff and know if it is right
- My role: reviewer and taste-keeper

rust-gpu + claspr — uncharted

- Graphics SPIR-V to OpenCL SPIR-V — Rust on Aurora
- I am an OpenCL expert; not a Rust or SPIR-V expert
- Nothing comparable exists in the training corpus
- My role: oracle designer

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CCS IN SIXTY SECONDS

Autotuning configuration spaces, shared across languages

- C99 library: configuration spaces, objective spaces, tuners, ask/tell
- Numerical, categorical, ordinal, discrete and string parameters
- Conditions, forbidden clauses, expressions, feature contexts
- JSON and binary serialization; Python and Ruby bindings
- So a tuner written in one language can be driven from a runtime in another
- About 53,000 lines of C — and I know all of them

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RUST-GPU + CLASPR IN SIXTY SECONDS

Rust on Intel GPUs, and a single-source programming model

- rust-gpu compiles Rust to SPIR-V — but targeted Vulkan only, so no Aurora
- We added the OpenCL Kernel execution model: Physical64 addressing, OpenCL.std intrinsics, native CL vector types
- claspr is the host layer: single-source proc-macros, typed launches, type-state buffer safety
- One async chain composes multi-stage GPU work into a single event graph

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PART 1
WHAT ACTUALLY LANDED

WHAT LANDED

Six weeks on CCS; four months on rust-gpu, claspr and pocl

CCS

- 99 pull requests, #24 through #122 — 96 merged, 3 closed
- Feb 26 → Apr 7, about 16 merged PRs per week
- Sanitizer and coverage jobs added to CI
- Every PR single-topic, read and merged by hand

rust-gpu · claspr · pocl

- rust-gpu: +24,262 / −384 across 765 files; OpenCL 1.2 and 2.0 targets
- claspr: 466 commits, ~49K lines of Rust, 417 tests
- Green on three OpenCL runtimes: pocl, rusticl, Intel NEO
- pocl upstream: 4 PRs merged, 2 issues filed and fixed

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WHAT THE CORRECTNESS WORK LOOKED LIKE

Real defects, in both directions — found by tools, not by reading

CCS — via UBSan and gcov

- Function-pointer type mismatches across the ops hierarchy — textbook C UB
- Null pointer passed to memcpy when the size is zero
- JSON integers silently losing precision past $\pm 2^{53}$
- Six real bugs in the Ruby and Python bindings

rust-gpu — via difftests

- A genuine miscompile: OpIAdd with mismatched operand widths
- Signedness stripped at emission, kept internally for dispatch
- spirv-opt segfaults on valid SPIR-V — now run in a forked child
- Kernel argument order preserved via a side-channel decoration

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UPSTREAM IN SOMEONE ELSE'S RUNTIME

Bugs fixed in three independent OpenCL implementations

What got fixed

- pocl: four merged PRs, including two genuine data races
- Mesa / rusticl: anonymous functions segfaulted the compiler
- Intel compute runtime: struct-by-value wrong as a kernel argument
- Found by our workloads, minimized, then fixed by their maintainers

What the maintainer caught

- Changes requested — seven inline comments, none of them logic errors
- All seven were naming, redundancy or misplaced checks: taste
- Two flagged the agent's own comments contradicting each other
- It could not see that. A human reviewer could, in one pass.

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PART 2
HOW THE HUMAN HAS TO WORK

DIRECTING VERSUS DIALOGUE

Same tool, same model — two completely different jobs for me

Directing (CCS)

- Branch per task, pull request per task, single topic each
- I am the reviewer: every PR read and merged by hand
- Rebase on upstream after every merge
- Short PRs keep review cost below the value delivered

Dialogue (rust-gpu + claspr)

- Start from a reproducer, not a specification
- Argue the design first — I bring the OpenCL semantics
- Spike first, land second; the spike is the design document
- I cannot check the code, so I check the behaviour

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HOW I VERIFIED

When you cannot read the diff, the oracle has to move outside you

Familiar: verify by reading

- I read the diff, and I know the idioms it should have used
- Backed by valgrind, sanitizers, coverage and distcheck
- Review is cheap because my knowledge is the oracle
- Risk: habits leak past me while I read for correctness

Uncharted: verify by construction

- Three OpenCL runtimes as a differential oracle — they must agree
- Four-way difftests: host and device, Rust and OpenCL, byte-for-byte
- CPU golden references, checked bit-exact
- Type-state buffers plus 24 compile-fail fixtures

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VERIFICATION AS ARCHITECTURE

Four things that caught real bugs the agent could not see

Differential runtimes

Run on pocl, rusticl and Intel NEO. They disagree — and the disagreement is the test.

Sanitizers and tracers

UBSan found the C undefined behaviour. An API tracer found 16 context allocations and 0 releases.

Bisect and repetition

Six runs per commit to find a race. 40/40, 60/60, bit-exact 5/5 to believe it is fixed.

Compile-time gates

Type-state markers and golden-stderr fixtures. Invariants the agent cannot quietly regress.

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MAKING THE ORACLE STRUCTURAL

One Rust function: the kernel runs it, the host validates with it

Code

```
#[claspr::device]
mod gpu {
    /// Pure Rust — called from the kernel AND from the host.
    pub fn collatz(mut n: u32) -> Option<u32> {
        let mut i = 0;
        while n != 1 { n = if n%2 == 0 {n/2} else {3*n+1}; i += 1; }
        Some(i)
    }

    #[claspr::kernel]
    pub fn collatz_kernel(
        #[spirv(global_invocation_id)] id: glam::USizeVec3,
        #[spirv(cross_workgroup)] data: &mut [u32],
    ) { data[id.x] = collatz(data[id.x]).unwrap_or(u32::MAX); }
}

let dev = DeviceSlice::from_slice(&ctx, &h)?;
let d = kernels.collatz_kernel([N], dev).wait()?;
d.read(&mut h).wait()?;

// Validate device output against the SAME function on the host.
let ok = (1..=N as u32).zip(&h)
    .all(|(i, &n)| n == gpu::collatz(i).unwrap_or(u32::MAX));
assert!(ok);
```

Why this matters here

- One function, two compilers: rust-gpu to SPIR-V, cargo to the host
- No FFI mock, no hand-written reference to drift out of sync
- The validator is the implementation — it cannot disagree by accident
- Build the oracle in once, instead of trusting every change

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PART 3
WHERE IT BROKE, AND WHAT TO DO ABOUT IT

PITFALLS DIFFER BY TERRAIN

Familiar code fails quietly; uncharted code fails late

On familiar code

- Silently swapped clang-format for clang-format-17 — CI stayed green
- Defaults to laziness on cleanup paths and overflow checks
- Forgets project idioms after every compaction
  - — Mitigation: short PRs and real diff review

On uncharted code

- Context exhaustion: eight auto-compactions in one session
- “Cleaned up” spike code that was intentional design documentation
- An edit anchor silently deleted a CI step
  - — Mitigation: end sessions early, pin toolchains

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THE BUG A FRIENDLY RUNTIME HIDES

Your development box's forgiveness is a liability

- claspr kept a buffer permanently mapped and also passed it to kernels
  - Straight undefined behaviour per the OpenCL spec — no escape hatch
- pocl is permissive, so it worked. For weeks. Every test green.
- rusticl is correct, so it failed: host wrote 6, kernel sees 0
- pocl is the outlier here, not rusticl — the forgiving runtime taught us wrong
- A second implementation is not redundancy. It is why we found this at all.

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THE AGENT MISDIAGNOSED ITS OWN BUG

Its narrative was confident and wrong; the tools were right

- A simulation diverged. The agent declared the race pre-existing.
- Bisect, six runs per commit: green at a049ad4, first red at 236d9c0
  - Its own commit — and the “clean baseline” it cited was inside the work it was defending
- A CPU golden broke the symmetry: equality cannot say which side is wrong
- The API trace gave the one-line proof: wait=NONE became wait=[3]
- Fixed — then bit-exact, five runs out of five

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CONTEXT IS A RESOURCE — SO OPTIMIZE IT

A performance-engineering reflex, turned on the agent itself

The observation

- “A sub-agent burns ~200K tokens before it can make ANY change”
- The architecture had grown too complex to hold in a context window
- In 2025 I blamed the models partly on rust-gpu being spaghetti
- In 2026 I stopped calling that an excuse and measured it

The optimization

- Profiled with rust-code-analysis to rank functions by complexity
- Worst function: cognitive 39 → 7
- Largest module: 11,023 → 3,674 lines, down 67%
- Behaviour-preserving, verified bit-identical on three runtimes

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WHAT ACTUALLY WORKS

Four gates, each earned by getting it wrong first

Diff review is the gate

Not CI-green. Toolchain swaps, scope creep and dependency drift all pass CI comfortably.

Pin your toolchain

Formatter version, runtime build, which driver actually loads. Verify what is loaded, not what is installed.

Get a second opinion

A second implementation, a second model, a sanitizer, a tracer. Never the agent re-reading its own work.

End the session

Every compaction costs quality. Long rolling sessions feel productive and quietly degrade.

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TAKEAWAY

You cannot let an agent run blindly and trust its work

- It produced real systems software — merged data-race fixes in an OpenCL runtime
- It also shipped spec-UB for weeks and misdiagnosed its own race
- Every bug here was caught by a sanitizer, a tracer, a bisect, a second implementation, or a human
  - Never once by the agent re-reading its own code
- Good news: you are spending two weeks learning exactly that toolbox
- What stays human: choosing the oracle, and reading the spec adversarially

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